

Answers

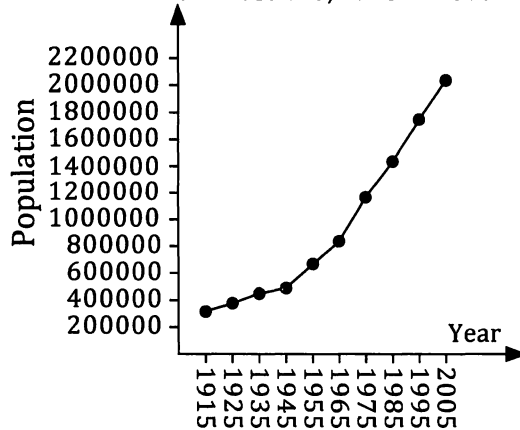
Exercise 1A. Page 26.

1. (a) No, the increase cannot be attributed to the natural increase in the population. Numbers are given *per 100 persons of working age*, thus the increase cannot simply be attributed to increase in population.



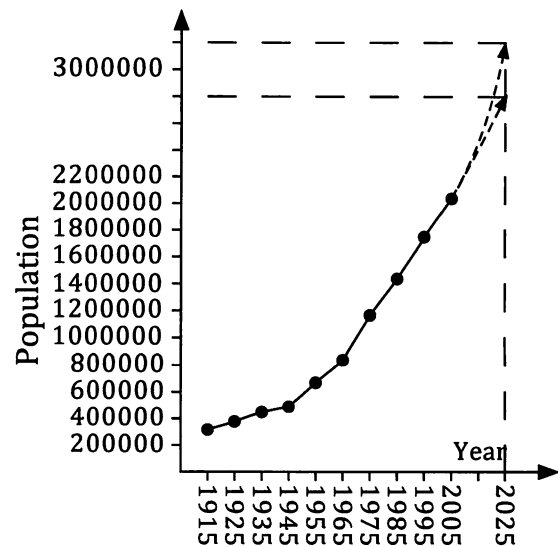
- (c) The number of persons over age 65 per 100 persons of working age shows an increasing trend with time. The increase is steady at approximately 2 more people per 100, for each ten years from 1971 to 2011, and then increases to about 5 more people per 100, for each ten years from 2011 to 2031.

2. (a) Line graph showing the population of Western Australia, 1915 to 2005.

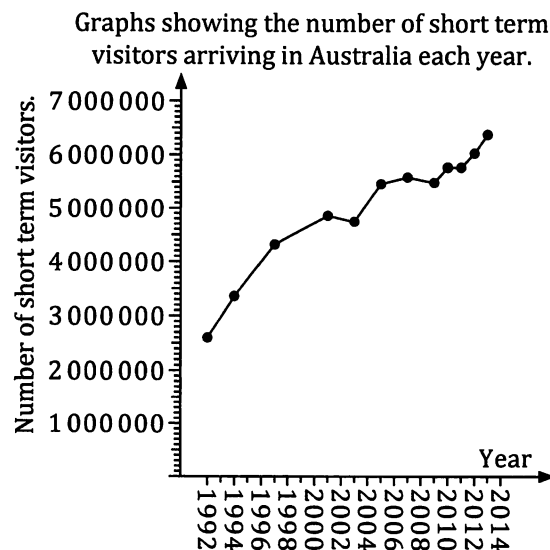


- (b) The population shows an increasing trend over the years from a population of just over 300 000 in 1915 to just over 2 million in 2005. Earlier 10 year intervals in this time show an increase in population of about 60 000 people per ten years. Later 10 year intervals show an increase in population of about 300 000 people per ten years.
- (c) The figures suggest that the population of Western Australia in 1950 was approximately 580 000. This prediction involves interpolating between two known points (1945, 490 000) and (1955, 669 000). Hence the prediction should be very reliable.

- (d) Continuing the trend suggests a population in 2025 of about 3 million (see graph on the right which shows two possible continuations to the trend). However extrapolation is involved as we are going beyond the known data points to make our prediction. Continuing the line as shown is an attempt to predict what will happen, but a lot could change between the 2005 figure and 2025 so with this degree of extrapolation involved the prediction should not be viewed as being particularly reliable. If we had to predict in a range, say 2.8 million to 3.2 million.



3. (a)



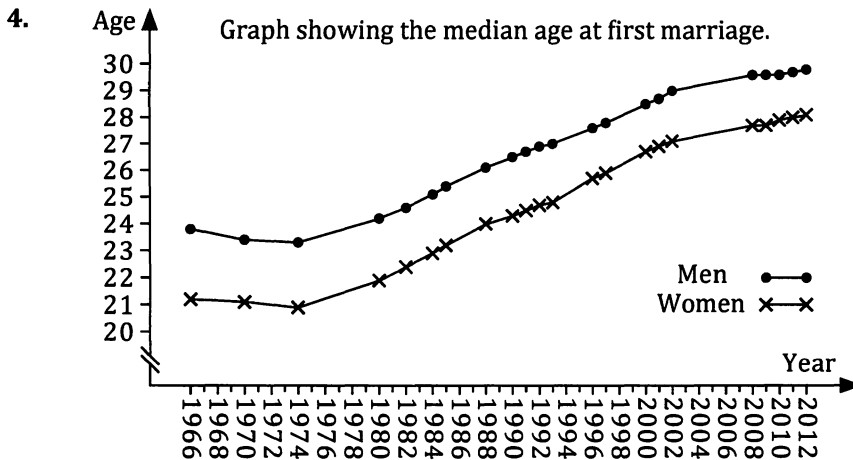
The number of short term visitors shows an increasing trend with time.

The increase seems reasonably steady from 1992-1997, then the rate of increase slows, with numbers actually declining in 2003 before picking up again. From 2005 to 2010 numbers of visitors are reasonably steady before increasing again from 2011 to 2013.

- (b) The graph suggest visitor numbers of approximately 4 600 000 for 1999. This prediction involves interpolation so it should be reliable.

The flattening of the graph around 2005 to 2010 makes extrapolation to 2020 difficult to do with any confidence. The 2013 figure perhaps suggests the upward trend continues so a prediction of approximately 7 500 000 for 2020 might be reasonable.

We are not extrapolating that far beyond the known values but extrapolation is involved so our prediction could not be relied upon with any great confidence.



Written response not included here.

Compare your response with those of others in your class and discuss with your teacher.

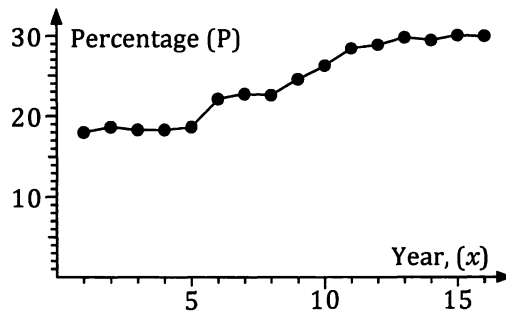
5. $N = 494Y - 917000$

6. (a) Number of properties managed = $290 + 11.4t$

(b) 655

7. (a) $P = 0.8x + 17.2$

(b)

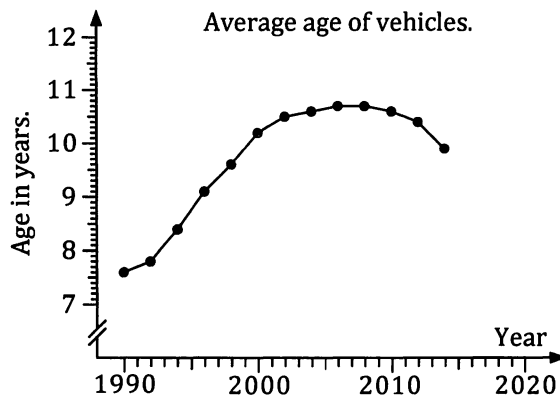


Over the sixteen years the percentages of births delivered by caesarean section show an increasing trend with most of the increase being from year 5 to year 11.

For the first five years the percentage remains level at around 18%.

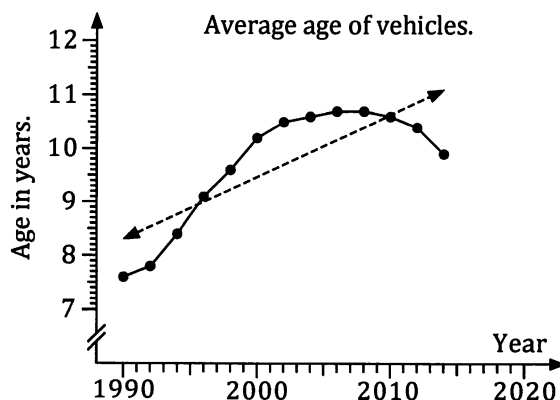
For the last four years the percentage remains level at around 30%.

8. (a) Average age of vehicles.



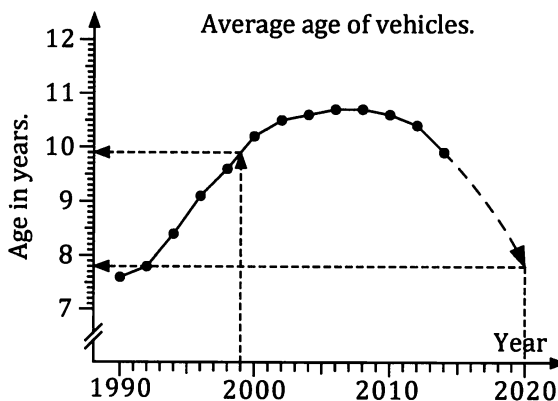
- (b) The data is not suitable for linear regression. The graph shows that the distribution of points does not follow a linear pattern, nor anything that could be well approximated by a linear pattern.

Predicted values given by the "best straight line" as determined using linear regression would overestimate ages for the early and late years, and underestimate ages for the middle years – see diagram below in which the least squares linear regression line is shown as a broken line.



- (c) Predicted average age of vehicle for 1999 would be about 9.9 years. We have data for 1998 and 2000 so our value for 1999 involves interpolation. It should therefore be a prediction that could be relied on with confidence.

We cannot predict with any great confidence what the average age of vehicles will be in 2020 as 2020 is well beyond the years plotted (i.e. extrapolation is involved). If decline in later years were to continue, the figure for 2020 might perhaps be around 7.8. No great confidence in reliability of this prediction though.



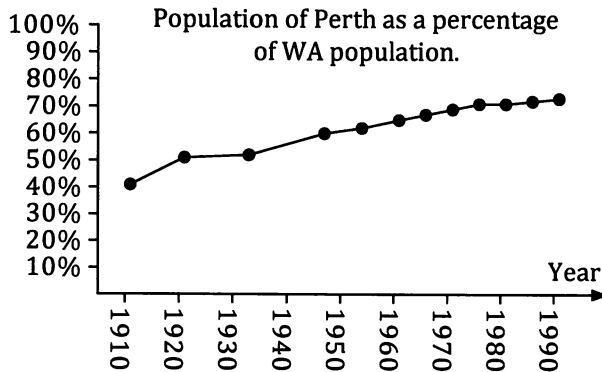
9. Viewing the graph of Y plotted against t confirms the unsuitability of linear regression.
 $Y = 4.69 \times t^{2.145}$.

Miscellaneous Exercise One. Page 32.

1. \$2316.50
2. \$3276
3. A: $y = x$ B: $y = 2x + 1$ C: $y = 0.5x + 1$ D: $x = 3$ E: $y = 2$ F: $y = -2x - 3$
4. (a) $T_1 = 25$ $T_2 = 27$ $T_3 = 29$ $T_4 = 31$ $T_5 = 33$
 (b) $T_1 = 32$ $T_2 = 30.5$ $T_3 = 29$ $T_4 = 27.5$ $T_5 = 26$
 (c) $T_1 = 64$ $T_2 = 96$ $T_3 = 144$ $T_4 = 216$ $T_5 = 324$
 (d) $T_1 = 4000$ $T_2 = 2000$ $T_3 = 1000$ $T_4 = 500$ $T_5 = 250$
 (e) $T_1 = 5$ $T_2 = 17$ $T_3 = 41$ $T_4 = 89$ $T_5 = 185$
 (f) $T_1 = 16$ $T_2 = 20$ $T_3 = 26$ $T_4 = 35$ $T_5 = 48.5$

5.

Percentage of WA population living in Perth												
Yr	1911	1921	1933	1947	1954	1961	1966	1971	1976	1981	1986	1991
%	41%	51%	52%	60%	62%	65%	67%	68%	71%	71%	72%	73%



Perth population, as % of WA population, shows an increasing trend.

This Increase "levels off" in later years but the percentage is still increasing.

(b) (i) For 1940 approx 56%.

1940 is between years for which the population is known so interpolation is involved.

Estimate should be reliable.

(ii) For 2020 approx 80%.

Extrapolation is involved and 2020 is a long way beyond known values. Estimate may not be too reliable.

(Note: Linear regression gives 86% but not a particularly suitable model.)

Exercise 2A. Page 37.

1. 3 point

2. 4 point

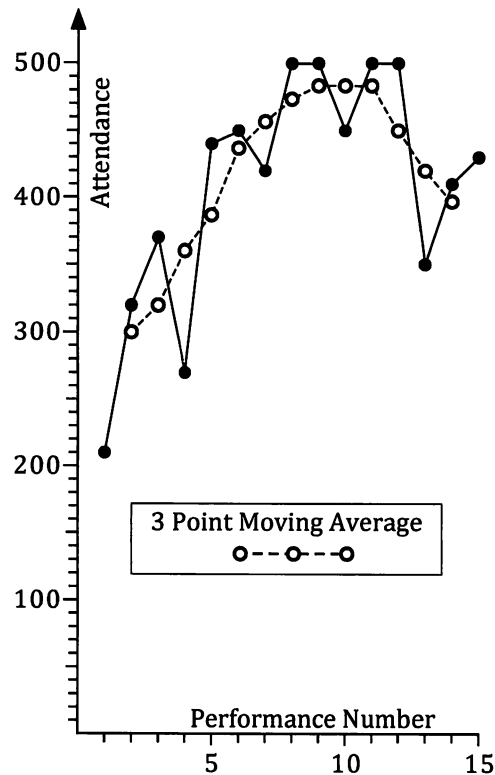
3. 4 point

4. 3 point

5. 5 point

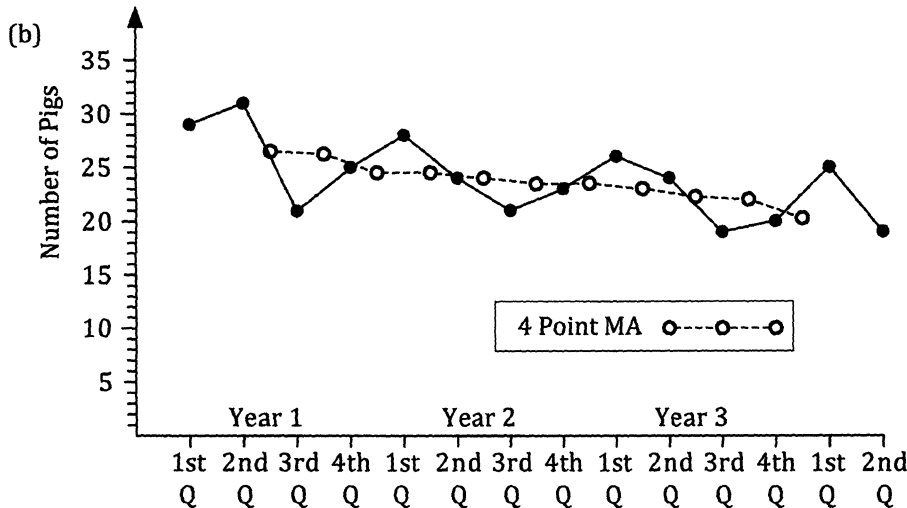
6. 3 point

7. (a) and (b) See graph:



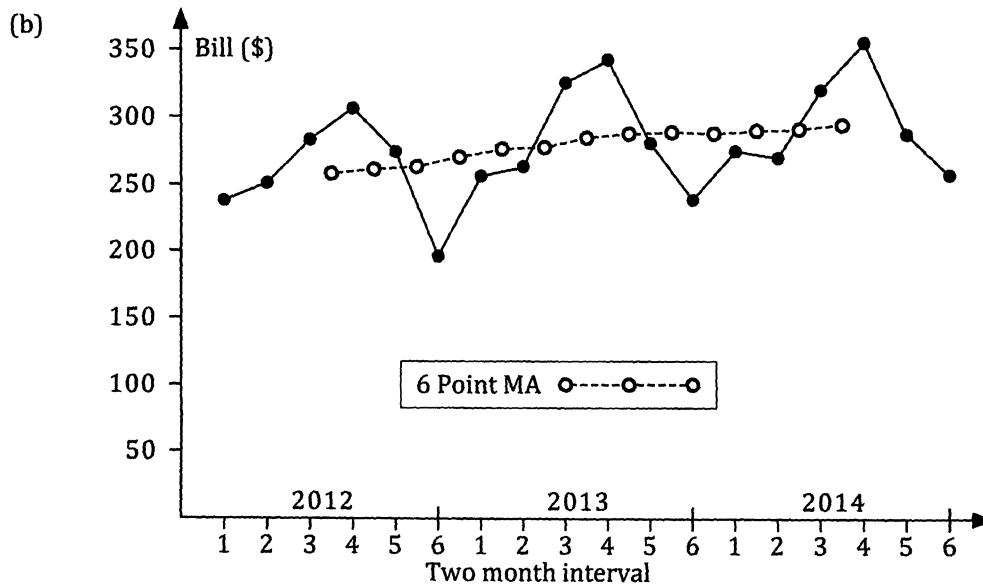
- (c) Many factors such as cost, venue and actor availability, would need to be considered before we could decide if a sixth week would be "a good idea". However, attendance figures suggest a 6th week could possibly have given attendance similar to week 1 so on that basis alone, and if the attendance of week 1 was considered acceptable, a 6th week could have been a good idea.

8. (a) Four point moving averages are:
26.5, 26.25, 24.5, 24.5, 24, 23.5, 23.5, 23, 22.25, 22, 20.75.



(c) Decreasing.

9. (a) A = 263, B = 270, C = 276, D = 277, E = 256.



(c) Increasing.

Exercise 2B. Page 43.

- Consumption of soft drink could well be seasonal in nature. Hence, with the data involving consumption per quarter year a 4 point average would be most suitable.
 - A = 6940, B = 6892, C = 6788, D = 6763
E = 5511, F = 6638, G = 6662
- A = 176.5, B = 180, C = 179, D = 182, E = 183, F = 184
G = 181, H = 190, I = 189, J = 191.5

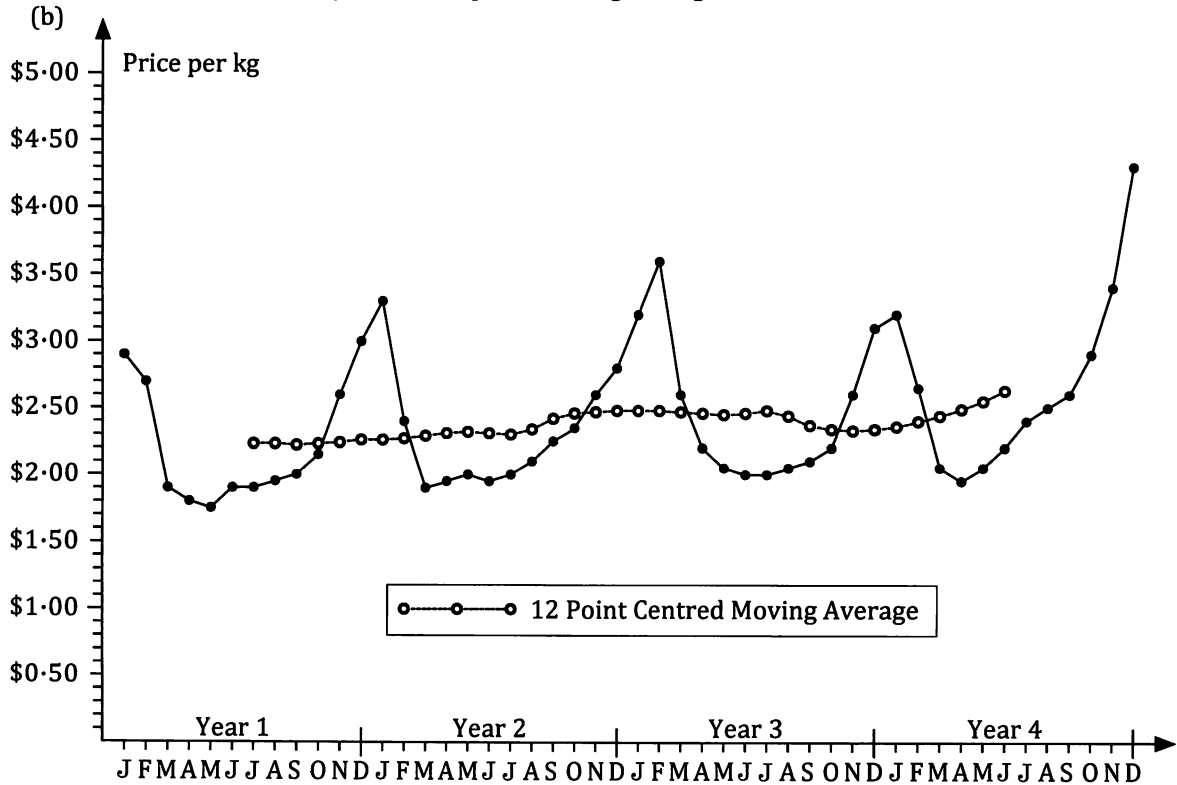
3.

Year 1				Year 2			
Summer	Autumn	Winter	Spring	Summer	Autumn	Winter	Spring
1228	364	640	1220	1436	276	752	1132

4 Point MAs	863	915	893	921	899
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4 Point CMAs	889	904	907	910
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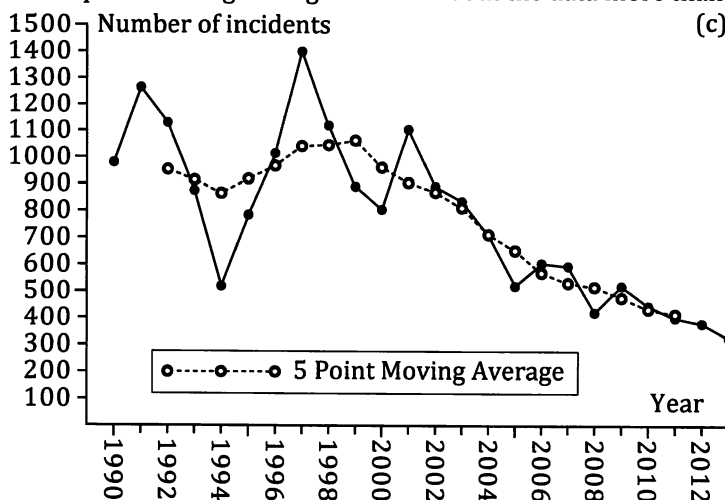
4. (a) We would expect the production of fruit to be dependent on the weather and hence follow a seasonal pattern. Hence determining a seasonal moving average makes sense and with data collected on a monthly basis a 12 point moving average is sensible.



- (c) Underlying trend shows a slight increase over time.

5. (a) The 5 point moving average would smooth the data more than the 3 point moving average.

(b) Number of incidents



- (c) In the years under consideration there is an initial downward trend followed by an upward trend, but then from about the year 2000 the trend has been decreasing.

Hence the expectation mentioned at the beginning of the question, i.e. that with increasing population we would expect the number of criminal incidents to increase, was not the case for this category of crime.

(From 1990 to 2013 the NSW population increased from 5.8 million to 7.4 million.)

Exercise 2C. Page 50.

- The seasonal index for the fourth quarter is 98%.
- The seasonal index for Wednesday is 94%.
- With no seasonal effect on sales the company should expect to sell roughly 13000 each month.
 - Predicted sales for each month would be as follows:

January	11 440	February	10 660	March	9 880
April	9 620	May	10 790	June	12 740
July	13 520	August	13 650	September	14 040
October	14 430	November	16 380	December	18 850
- The figures suggest that sales for the whole year will be 62 000.
For each of the remaining quarters the figures suggest sales will be:
2nd quarter 16 120, 3rd quarter 15 190, 4th quarter 13 330.

Week	Period of time	Attendance	Daily mean for the week.	Attendance as % of week's daily mean.
1	Friday	2 346	2 766	84·82%
	Saturday	3 143		113·63%
	Sunday	2 809		101·55%
2	Friday	2 572	2 929	87·81%
	Saturday	3 258		111·23%
	Sunday	2 957		100·96%
3	Friday	2 987	3 231	92·45%
	Saturday	3 500		108·33%
	Sunday	3 206		99·23%

	Friday	Saturday	Sunday
1st week	84·82%	113·63%	101·55%
2nd week	87·81%	111·23%	100·96%
3rd week	92·45%	108·33%	99·23%
Index (1 dp)	88·4%	111·1%	100·6%

Year	Period of time	Units sold	Quarterly mean for the year.	Units as % of year's quarterly mean.
1	1st quarter	54 000	52 650	102·56%
	2nd quarter	63 200		120·04%
	3rd quarter	45 200		85·85%
	4th quarter	48 200		91·55%
2	1st quarter	48 500	47 250	102·65%
	2nd quarter	56 700		120·00%
	3rd quarter	41 300		87·41%
	4th quarter	42 500		89·95%
3	1st quarter	51 800	49 675	104·28%
	2nd quarter	62 600		126·02%
	3rd quarter	40 500		81·53%
	4th quarter	43 800		88·17%

	1st quarter	2nd quarter	3rd quarter	4th quarter
1st year	102·56%	120·04%	85·85%	91·55%
2nd year	102·65%	120·00%	87·41%	89·95%
3rd year	104·28%	126·02%	81·53%	88·17%
Seasonal Index (1 dp)	103·2%	122·0%	84·9%	89·9%

7.

<i>n</i>	Week	Day	N ^o . of calls	5pt MA	Calls as % of daily average for week.
1	One	Mon	258	—	107.59%
2		Tue	231	—	96.33%
3		Wed	215	239.8	89.66%
4		Thur	248	235.6	103.42%
5		Fri	247	231.0	103.00%
6	Two	Mon	237	224.8	109.32%
7		Tue	208	221.6	95.94%
8		Wed	184	216.8	84.87%
9		Thur	232	215.8	107.01%
10		Fri	223	213.0	102.86%
11	Three	Mon	232	212.8	113.06%
12		Tue	194	208.2	94.54%
13		Wed	183	205.2	89.18%
14		Thur	209	—	101.85%
15		Fri	208	—	101.36%

	Week 1	Week 2	Week 3	Seasonal index
Monday	107.59%	109.32%	113.06%	109.99%
Tuesday	96.33%	95.94%	94.54%	95.60%
Wednesday	89.66%	84.87%	89.18%	87.90%
Thursday	103.42%	107.01%	101.85%	104.09%
Friday	103.00%	102.86%	101.36%	102.41%

Exercise 2D. Page 57.

(You may occasionally find that an answer you obtain varies slightly from the answer given here depending on when, and to what degree, rounding occurred.)

- \$22300 (= \$25400 ÷ 1.14 and then given to the nearest \$100).
- 173700 units (= 132000 ÷ 0.76 and then given to nearest 100).
- 69530 people.
 - 69940 people.
- The analysis predicts that 14750 units will be sold in January.
- The figures suggest that the actual weight of the fruit this summer will be 23680 kg.
- Seasonal index for Jan – Apr: 107.25%
 Seasonal index for May – Aug: 75.56%
 Seasonal index for Sept – Dec: 117.20%
 - | | | | |
|--------|-----------|-----------|------------|
| | Jan – Apr | May – Aug | Sept – Dec |
| Year 1 | 1.71 | 1.71 | 1.69 |
| Year 2 | 1.74 | 1.87 | 1.76 |
| Year 3 | 1.80 | 1.69 | 1.84 |
| Year 4 | 2.03 | 2.01 | 1.99 |

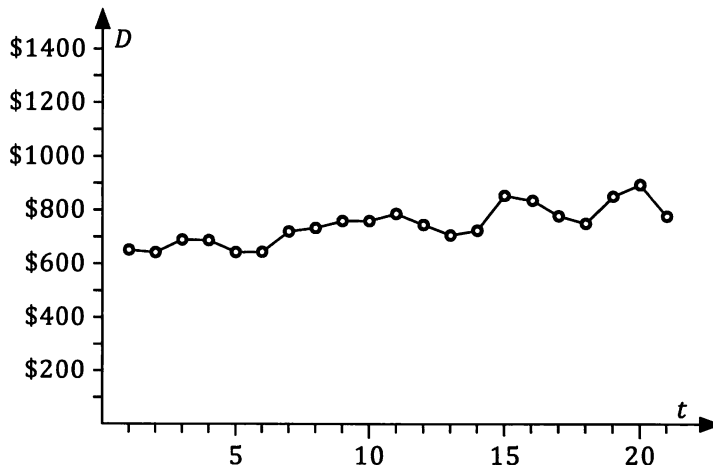
7. (a) Seasonal index for Friday: 0.6436 Seasonal index for Saturday: 1.0540
 Seasonal index for Sunday: 1.1555 Seasonal index for Monday: 1.1469

	Friday	Saturday	Sunday	Monday
Year 1	14290	14080	14070	14690
Year 2	15240	14980	14950	15520
Year 3	14680	14440	14890	14660
Year 4	15290	16030	15610	14600

8. (a)

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 1	\$652	\$643	\$690	\$688	\$644	\$645	\$721
Week 2	\$734	\$760	\$760	\$787	\$746	\$707	\$725
Week 3	\$855	\$837	\$779	\$751	\$852	\$895	\$778

- (b) Graph of deseasonalised data, D , plotted against t , as shown on the right, shows a reasonable linear trend. Hence linear regression is appropriate.



(c) $D = 9.64t + 639.1$

- (d) For Monday of week 4 ($t = 22$) the regression line predicts value for D of \$851.2.
 Hence prediction of real takings will be $\$851.2 \times 0.8242 = \702 , to the nearest dollar.
 For Wednesday of week 4 ($t = 24$) the regression line predicts value for D of \$870.5.
 Hence prediction of real takings will be $\$870.5 \times 0.5394 = \470 , to the nearest dollar.
 For Saturday of week 4 ($t = 27$) the regression line predicts value for D of \$899.4.
 Hence prediction of real takings will be $\$899.4 \times 1.4194 = \1277 , to the nearest dollar.

9. (a) Viewing the graph, not shown here, confirms reasonableness of using linear regression for (n, M) data.

(b) $M = 1.0994n + 26.031$

(c)

	Monday	Tuesday	Wednesday	Thursday	Friday
Week one	1.15132	0.82237	0.78947	0.92105	1.31579
Week two	1.07558	0.87209	0.78488	0.87209	1.39535
Week three	1.14428	0.79602	0.82090	0.87065	1.36816
Week four	1.04348	0.91304	0.86957	0.95652	1.21739
Mean (4dp)	1.1037	0.8509	0.8162	0.9051	1.3242

- (d) Missing figures are:
- | | Tuesday | Wednesday | Thursday |
|------------|---------|-----------|----------|
| Week one | 29 | 29 | 31 |
| Week two | 35 | 33 | 33 |
| Week three | 38 | 40 | 39 |
| Week four | 49 | 49 | 49 |

- (e) Predictions for week 5: Mon 54 Tues 43 Wed 42 Thur 47 Fri 71

- (f) If week five involved things like "pupil free days", public holidays, unusually severe weather making travel to school difficult, a flu epidemic in the area, school closure due to fire etc, our week five attendance predictions would not be reliable. (Also, we cannot assume that the trends of the first four weeks will necessarily continue even without these unusual events.)

10. (a)

Rescues as a percentage of annual mean. (As percentages and correct to 2 decimal places).			
	Oct/Nov	Dec/Jan	Feb/Mar
Season 1	56.10%	135.37%	108.54%
Season 2	47.37%	136.84%	115.79%
Season 3	51.43%	152.86%	95.71%
Season 4	53.13%	142.19%	104.69%

Seasonal indices (Nearest percent.)	52%	142%	106%
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	Oct/Nov	Dec/Jan	Feb/Mar
Season 1	88	78	84
Season 2	69	73	83
Season 3	69	75	63
Season 4	65	64	63

(c) Viewing the graph, not shown here, confirms reasonableness of using linear regression for (n, D) data.

Using integer values of D : $D = 85.83 - 2n$.

(Using more accurate values of D : $D = 86.05 - 2.003n$.)

n	Season	Months	Predicted number of rescues
13	5	Oct/Nov	31
14		Dec/Jan	82
15		Feb/Mar	59

Miscellaneous Exercise Two. Page 62.

1. \$247.50

2. 1

3. A seasonal index of 0.87 for February means that whatever it is that we are measuring for February tends to be 13% below the monthly average.

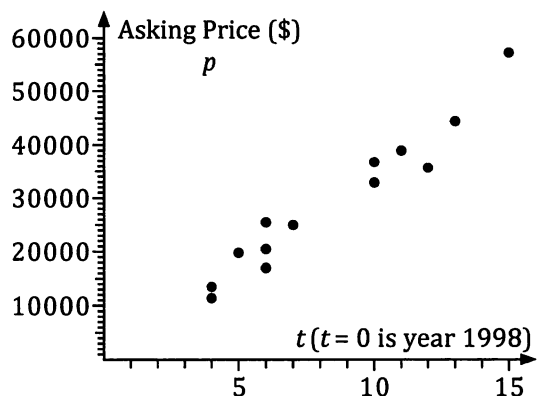
Season	Spring	Summer	Autumn	Winter
Deseasonalised data	1000	1552	1600	1875

5. The starting amount is \$1 500 000.

6. The scattergraph indicates that in general the more recently manufactured the vehicle the higher the asking price. Given the number of points involved, and their distribution, the relationship between year of manufacture and asking price could be well represented by a linear model.

$$p = 3561t - 697$$

Likely price for a 2007 model: \$31 350



7. (a) The given revenues have a total of \$3 760 000 and thus a quarterly average of \$940 000. The first quarter is 20% down on this quarterly average (compared to the 15% down the seasonal index suggests), the second quarter is 21% up (compared to the expected 25%), the third quarter is 25% up (compared to the expected 12%) and the fourth quarter is 26% down (compared to the expected 22%). Thus the third quarter revenue is a surprise when compared to what we might expect from the seasonal indices.
- (b) If all other factors remained the same then the third quarter increase in the revenue share above expectation could be as a result of the advertising campaign being successful.

8. (a)

t	Year	Quarter	Number sold in quarter.	Number sold as percentage of quarterly mean for the year.	Number sold seasonally adjusted.
1	One	1st	1637	106.99%	1514
2		2nd	1489	97.32%	1537
3		3rd	1244	81.31%	1598
4		4th	1750	114.38%	1494
5	Two	1st	1405	108.16%	1299
6		2nd	1241	95.54%	1281
7		3rd	1012	77.91%	1300
8		4th	1538	118.40%	1313
9	Three	1st	1253	109.24%	1159
10		2nd	1121	97.73%	1157
11		3rd	852	74.28%	1095
12		4th	1362	118.74%	1162

	Year 1	Year 2	Year 3	Seasonal index
1st Quarter	106.99%	108.16%	109.24%	108.13%
2nd Quarter	97.32%	95.54%	97.73%	96.86%
3rd Quarter	81.31%	77.91%	74.28%	77.83%
4th Quarter	114.38%	118.40%	118.74%	117.17%

- (c) Using integer values for seasonally adjusted figures: $S = -43.885t + 1611$.
 (Using more accurate values for seasonally adjusted figures: $S = -43.887t + 1611.1$.)
- (d) Predicted number sold for year 4 are:
 1st quarter 1125, 2nd quarter 965, 3rd quarter 742, 4th quarter 1065.

Exercise 3A. Page 68.

- \$2720
- 5 years
- 12.5%
- \$2400
- 7.4%
- \$7700
- \$6347.50 to the nearest cent.
- \$9609.32 to the nearest cent.
- Annual compounding gives interest of \$131 23.85, to the nearest cent.
Monthly compounding would give \$400.61 more interest, to the nearest cent.
- Annual compounding gives interest of \$463 570.00, to the nearest cent.
Daily compounding would give \$26 568.34 more interest, to the nearest cent.
- \$6965.59

12. It would take approximately 11.64 years for the investment to double in value. I.e. approximately 11 years and 8 months.
13. It would take almost 20 compoundings (19.89), i.e. just under 10 years (9.945 years).

	1 st term	2 nd term	3 rd term	4 th term	5 th term	15 th term
14.	5,	8,	11,	14,	17.	47.
15.	5,	2,	-1,	-4,	-7.	-37.
16.	-10,	-5,	0,	5,	10.	60.
17.	12.5,	15,	17.5,	20,	22.5.	47.5.
18.	0.25,	0.5,	1,	2,	4.	4096.
19.	20 480,	30 720,	46 080,	69 120,	103 680.	5 978 711.25.
20.	2 621 440,	1 310 720,	655 360,	327 680,	163 840.	160.
21.	2 ³⁰ ,	2 ²⁹ ,	2 ²⁸ ,	2 ²⁷ ,	2 ²⁶ .	2 ¹⁶ .
22.	5,	13,	29,	61,	125.	131 069.
23.	\$1 000,	\$1 300,	\$1 660,	\$2 092,	\$2 610.40.	\$18 758.78.
24.	\$5 000,	\$5 000,	\$5 000,	\$5 000,	\$5 000.	\$5 000.
25.	12,	23,	34,	45,	56.	166.
26.	819 200,	819 200,	819 200,	819 200,	819 200.	819 200.
27.	1,	1,	2,	3,	5.	610.
28.	2,	3,	8,	19,	46.	309 268.
29.	5,	-5,	-5,	5,	5.	-5.
30.	5,	-3,	-1,	3,	5.	-1.

Exercise 3B. Page 71.

Percentages given rounded to three decimal places where rounding is necessary.

1.	Compounding frequency.	Nominal annual interest rate (%).	Effective annual interest rate (%).
	Annual	4%	4%
	Six monthly	4%	4.04%
	Quarterly	4%	4.060%
	Monthly	4%	4.074%
	Weekly	4%	4.079%
	Daily	4%	4.081%

2.	Compounding frequency.	Nominal annual interest rate (%).	Effective annual interest rate (%).
	Annual	8%	8%
	Six monthly	8%	8.16%
	Quarterly	8%	8.243%
	Monthly	8%	8.300%
	Weekly	8%	8.322%
	Daily	8%	8.328%

3. The effective annual interest rates for 8% are not simply double the rates for 4%.
Doubling the value of "i" in the formula $\left(1 + \frac{i}{n}\right)^n - 1$ will not simply double the answer.

Exercise 3C. Page 75.

- $T_{n+1} = 1.08 \times T_n + \200 \$11 533.01 is in the account at the end of ten years.
- (a) $T_{n+1} = 1.005 \times T_n + \100 (b) Final value of account is \$9817.01. ($= T_{36} - \100)
- \$22 474.58 (nearest cent). 4. \$4 780.04 (nearest cent).
- \$14 773.77 (nearest cent). 6. (a) 10.164% (correct to three decimal places).
(b) 7.590% (correct to three decimal places).

Exercise 3D. Page 77.

- (a) End of first year \$195 000 (b) End of first year \$187 000
End of second year \$170 000 End of second year \$158 950
End of third year \$145 000 End of third year \$135 107.50
- (a) First year depreciation is \$10 000 (b) First year depreciation is \$16 000
Second year depreciation is \$10 000 Second year depreciation is \$12 800
Third year depreciation is \$10 000 Third year depreciation is \$10 240
- (a) \$23 000 (b) \$16 000 (c) \$9 000

Exercise 3E. Page 79.

- A: \$182.7389691 B: \$182.74 C: 12 D: \$4 000 E: \$0
F: 2 G: 9 H: monthly I: 12
- A = 5 359, B = 3 894.90, C = 512.83, D = 564.11.
- A = 1.015, B = 1043.86, C = 180.64.
- \$49 700, \$49 397, \$49 090.97, \$48 781.88 (nearest cent).
- (a) At the end of year five and immediately after his regular monthly repayment, Hisham still owes \$15 016.34 on the loan.
(b) Hisham would owe \$12 974.59 at the end of the 5 years (after his repayment for the month).
- (a) Pays off the loan at the end of the 70th month (and the 70th repayment is less than \$250).
(b) Pays off the loan at the end of the 56th month (and the 56th repayment is less than \$290).
(c) Pays off the loan at the end of the 48th month (and the 48th repayment is less than \$290).
- The constant monthly repayment needs to be \$443.21, to the nearest cent.
- The constant monthly repayment needs to be \$717.35, to the nearest cent.
- The constant monthly repayment needs to be \$1 972.11, to the nearest cent.
- The loan would be paid off in exactly two years by paying \$188.29 per month for 24 months (actually a few cents more on the very last repayment to exactly pay off the outstanding amount).
If instead the monthly interest rate was 1.2% per month the monthly repayments would need to be \$192.81 (now a few cents less on the very last repayment).

Exercise 3F. Page 84.

- Loan of \$514 214.53 + deposit of \$17 000.
Hence, to the nearest \$1 000, Fran and Michael can afford a house costing \$531 000.
- Loan of \$305 688.95 + deposit of \$20 000.
Hence, to the nearest \$1 000, Peta and Peter can afford a house costing \$326 000.
- Loan of \$441 419.57 + deposit of \$35 000.
Hence, to the nearest \$1 000, Rania and Umah can afford a house costing \$476 000.
- Loan of \$234 000.19 + deposit of \$450 000.
Hence, to the nearest \$1 000, Hue and James can afford a house costing \$684 000.
- Loan of \$364 609.29 + deposit of \$30 000.
Hence, to the nearest \$1 000, Kirra can afford a house costing \$395 000.
- \$3 575.13 (Note: Round *down* from \$3 575.13092. Final payment will be a little *over* \$3 575.13.)
\$622 540 (nearest \$10) Note: $300 \times \$3 575.13 - \$450 000$ gives \$622 539.
Allowing for larger 300th payment gives \$622 539.93.
Different rounding regime, eg on some online amortization calculators, gives \$622 539.38.
- \$2 991.93 (Note: Round *up* from \$2 991.927336. Final payment will be a little *under* \$2 991.93.)
\$718 060 (nearest \$10) Note: $240 \times \$2 991.93$ gives \$718 063.20.
Allowing for reduced 240th payment gives \$718 061.78
Different rounding regime, eg on some online amortization calculators, gives \$718 061.84.
Paying \$100 more per month would pay off the loan approximately 16 and a half months earlier.

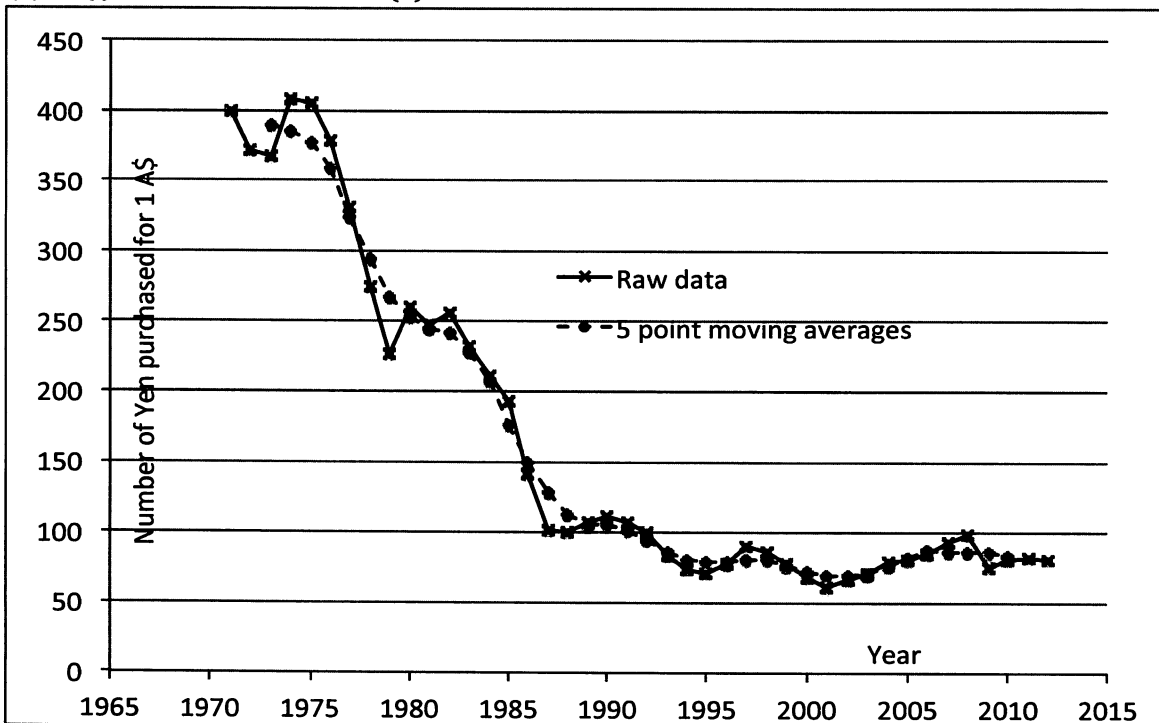
Miscellaneous Exercise Three. Page 84.

1. (a) \$11754.62
(b) \$11887.58
(c) \$11918.77
2. A seasonal index of 1.08 for Autumn means that whatever it is that the index is referring to, eg sales numbers, people moving interstate, unemployment figures, etc, the measurements for Autumn tend to be 8% above the average for a season.

3.

Season	Mon	Tue	Wed	Thur	Fri	Sat
Deeasonalised data	123	113	113	125	117	133

4. (a) $T_1 = 7$ $T_2 = 19$ $T_3 = 31$ $T_4 = 43$ $T_5 = 55$
 (b) $T_1 = 100$ $T_2 = 85$ $T_3 = 70$ $T_4 = 55$ $T_5 = 40$
 (c) $T_1 = 5000$ $T_2 = 6000$ $T_3 = 7200$ $T_4 = 8640$ $T_5 = 10368$
 (d) $T_1 = 2000$ $T_2 = 200$ $T_3 = 20$ $T_4 = 2$ $T_5 = 0.2$
 (e) $T_1 = 4$ $T_2 = 11$ $T_3 = 25$ $T_4 = 53$ $T_5 = 109$
 (f) $T_1 = 200$ $T_2 = 296$ $T_3 = 440$ $T_4 = 656$ $T_5 = 980$
5. (a) D (b) B, C (c) A (d) B, D (e) B
6. (a) \$48620.70 (b) \$45810.01 (c) \$41460.41 (d) \$32257.88
7. (a) \$16483.73
 (b) 15.19 years, i.e. just over 15 years and 2 months.
 (c) 18 years.
 (d) (i) at least 11.3264% (ii) at least 10.7777%
- 8.



Comments summarising the data, and about today's exchange rate, are not included here. Compare your comments to those of others in your class.

Exercise 4A. Page 91.

1. If \$8000 (or more) is invested at 7.5% per annum then the interest earned will be $\$8000 \times 0.075$ i.e. \$600 (or more). Hence the interest each year will equal (or exceed) the award amount. The balance in the account will then remain at (or exceed) \$8000 thus allowing the award to continue "in perpetuity".
2. $1.02^4 = 1.08243216$. Thus 8% per annum, with quarterly compounding, increases an initial investment by 8.243216%.
If the initial investment is \$A we require 8.243216% of \$A to equal \$75000.
Thus $\$A = \909839.0725 , i.e. \$909840 rounded up to the next dollar.
The one off investment needs to be \$909840 (or more).
(Other approaches are possible, for example trial and adjustment.)
3. \$243199.29 (or more).
4. (a) Increase P.
(b) An increase in R.
(c) A decrease in A.
5. There will be \$436670 left in the account after the tenth withdrawal (nearest ten dollars).
Kelvin can withdraw \$50000 per year for 22 years. (Then, at end of the 23rd year, the withdrawal would be less than \$50000 and would close the account.)
Had the interest rate been 7.8% Kelvin would have been able to withdraw \$50000 per year for 45 years. (Then, at the end of the 46th year, the withdrawal would be less than \$50000 and would close the account.)
6. Julie will be able to withdraw \$50000 from the account for 26 years. (Then, at the end of the 27th year, the withdrawal would be less than \$50000 and would close the account.)
If instead Julie withdrew \$45000 per year she could do this for 34 years. (Then, at the end of the 35th year, the withdrawal would be less than \$45000 and would close the account.)
To maintain her standard of living it is likely that Julie will need an increasing amount to live on each year because of the likely rise in the cost of everything due to inflation. Hence she might be wiser to withdraw an amount which increases each year to allow for this inflation, even if it means having to withdraw less than the \$50000 in the early years.
7. \$352941
8. \$575251
9. \$358663
10. \$454576
11. (a) and (b) Nineteen annual payments of \$30000 and then at the end of 20 years the final payment will be \$5214.89 ($= \4851.06×1.075).
12. (a) and (b) One hundred and seventy three monthly payments of \$4000 and then at the end of 174th month the final payment will be \$2534.41 ($= \2524.31×1.004).
13. (a) and (b) \$381236
14. (a) and (b) \$46732
15. (a) Immediately after the third withdrawal the balance in the account will be \$277722.60.
(b) Immediately after the 14th withdrawal the balance is \$26395.75. Hence at the end of the 15th year, by withdrawing \$28507.41 ($= \26395.75×1.08) the balance will be reduced to zero.
16. (a) Immediately after the eighth withdrawal the balance in the account will be \$149506.05.
(b) Immediately after the 13th withdrawal the balance is \$575.01 (\$575.009). Hence could simply add this to the payment at the end of the 13th year and reduce the balance to zero or could continue for another year and at the end of the 14th year, by withdrawing \$618.13 ($= \575.009×1.075) the balance will be reduced to zero then.

Exercise 4B. Page 97.

1. \$259162.18
2. \$567873
3. \$333114.12
4. \$565537.24
5. Fifty seven quarterly payments of \$15000 followed by a final payment of \$9924.23 which closes the account.

Computer spreadsheet or graphic calculator activity. Page 98.

Answers not given here. Compare your answers with those of others in your class.

Miscellaneous Exercise Four. Page 99.

1. \$2448.90
2. The 855 is the gradient of the line of best fit. It informs us that over the ten years under consideration the number of people attending the music festival increased at approximately 855 people per year.
3. (a) 6.136%
(b) 6.168%
(c) 6.183%
4. \$4267.45
5. (a) \$4300
(b) 6.5%
(c) \$6274.31
(d) 20.795%
6. The initial investment was \$1600, at a fixed monthly interest rate of 1.5% with a fixed monthly deposit of \$250.
7. (a) A: 0.7570 B: 0.8033 C: 0.7778 D: 0.7991
E: 160 F: 0.7625 G: 1.175

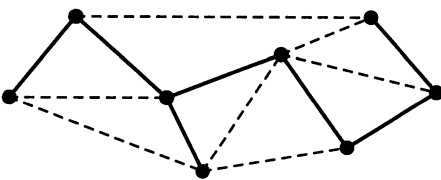
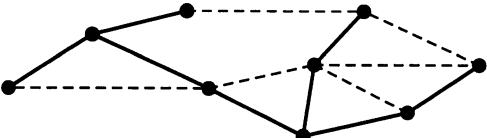
(b)

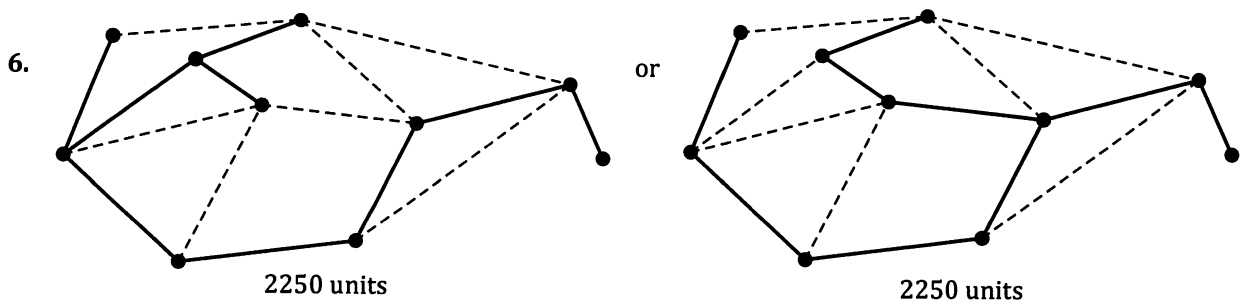
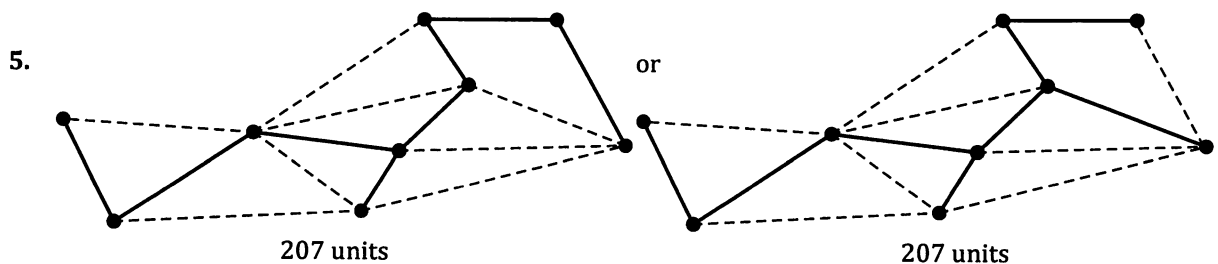
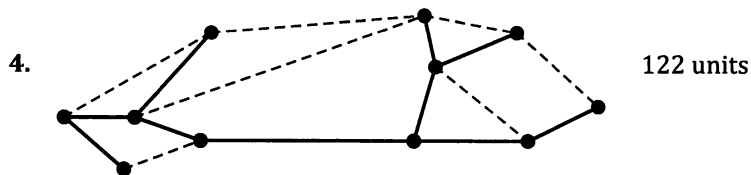
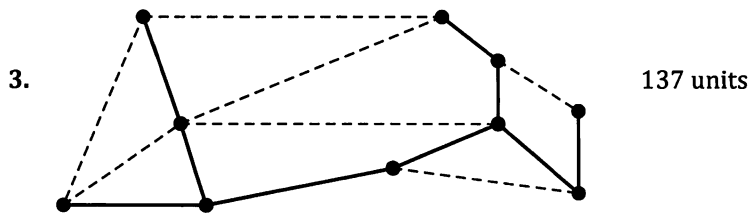
Calculation of Seasonal Indices.						
	2010	2011	2012	2013	2014	Seasonal Index (3dp)
1st 4 Months	1.1190	1.0233	1.1264	0.9953	1.0625	1.065
2nd 4 Months	0.7570	0.8033	0.7778	0.7991	0.7625	0.780
3rd 4 Months	1.1241	1.1733	1.0958	1.2056	1.1750	1.155

Seasonally adjusted sales figures, S . (Nearest integer.)					
	2010	2011	2012	2013	2014
First 4 Months	415	288	276	200	160
Second 4 Months	383	309	260	219	156
Third 4 Months	384	305	248	223	163

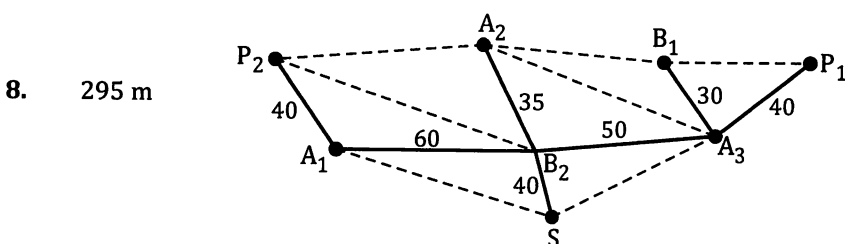
- (c) Using integer values for S : $S = -17.91t + 409.2$
(Using more accurate values for S : $S = -17.919t + 409.36$)
- (d) For the first 4 months of 2015 predicted sales are 131 (nearest integer).
For the second 4 months of 2015 predicted sales are 82 (nearest integer).
For the third 4 months of 2015 predicted sales are 100 (nearest integer).

Exercise 5A. Page 104.

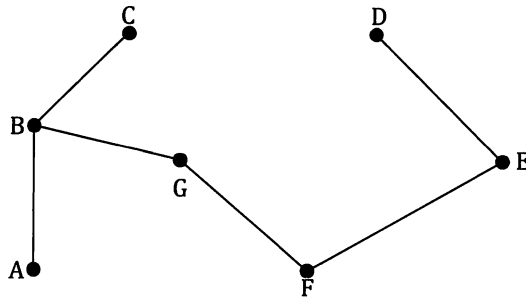
1.  188 units
2.  1920 units



7. Replace AG, GF, FC, CB, FE, ED with new piping. Close off AF, AC, AB, CE, CD, BD. New piping has a total length of 325 m.



9. (a)



(b) The total length of tunnelling that will be closed off to visitors is 445 metres.

10. (a) Graph showing roads and distances not given here. (b) 1890 km

Exercise 5B. Page 110.

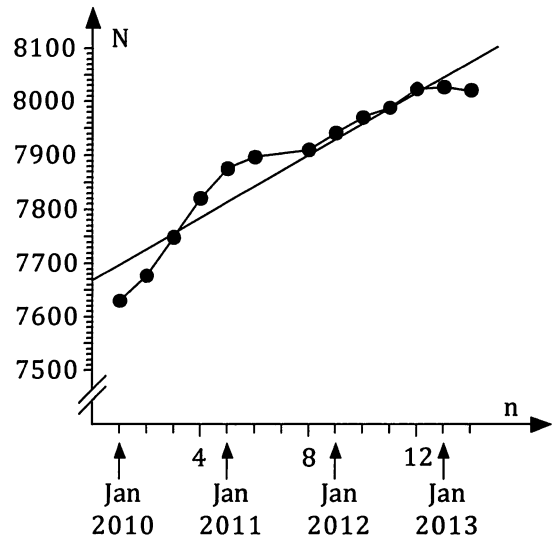
- | | |
|----------------|-----------------------------------|
| 1. 76 units | 2. 129 units |
| 3. 1 680 units | 4. 19.7 units |
| 5. 2 520 units | 6. 25.8 units, AB, BF, CF, DE, EF |
| 7. 21.1 km | 8. \$670 |

9. (a) 921 km (b) Harlow - Raine, Raine - Lewis, Wayley - Gatley, Raine - Fitch, Wayley - Olber, Fitch - Wayley

10. BF, CH, DF, EA, FE, GD, HG, IC, JC. 261 km.

Miscellaneous Exercise Five. Page 113.

1. (a) $N = 29.14n + 7668$
- (b) The value of a , 29.14, means that linear modelling of the data suggests that during the time period the data covers the number of people employed full time is increasing at the rate of approximately 29140 people per quarter.
- (c) July 2011 has an n value of 7. The line of regression gives $N = 7872$.
 Linear modelling gives a predicted employment figure for July 2011 of 7872 thousand.
 January 2015 has an n value of 21. The line of regression gives $N = 8280$.
 Linear modelling gives a predicted employment figure for January 2015 of 8280 thousand.
 We would expect the July 2011 prediction, which involves interpolation, to be more reliable than the January 2015 prediction, which involves extrapolation. The latter involves predicting beyond known data points and events could happen in the meantime making such predictions unreliable.
- (d) Viewing the data graphically, as shown on the right, suggests that linear modelling may be unwise. The overall trend seems more curved than straight. We could perhaps use two linear models but even then the apparent late downturn in employment suggests that extrapolating ahead of the given data based on a linear model could well be unwise. The placement of the linear regression line shows why the predicted figure for July 2011 was below the real figures on either side of it. Hence using a linear model for this data even made an interpolated prediction doubtful.

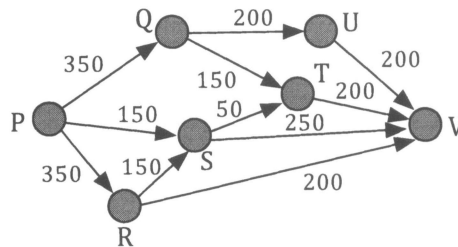


2. At 6% per annum, i.e. 0.5% per month, the initial \$350000 will earn interest of \$1750 by the end of the first month. Hence a withdrawal of \$1500 will leave more than \$350000 in the account for the next month. The account will continue to grow each month and the task set for the calculator, to find how many payments it takes to give a final balance of zero, is impossible. Hence the "Error" response.
3. $a = 67,$ $b = 76,$ $c = 80,$ $d = 71.5$ $e = 81,$ $f = 76,$
 $g = 80,$ $h = 78,$ $i = 69,$ $j = 82,$ $k = 80,$ $l = 86,$
 $m = 84,$ $n = 82,$ $o = 53.$
4. \$3634.60
5. (a) At the end of year six and immediately after her regular repayment, Amy still owes \$5636.26 on the loan.
 (b) To pay off the loan in exactly six years Amy's annual repayments need to be \$3264.44.

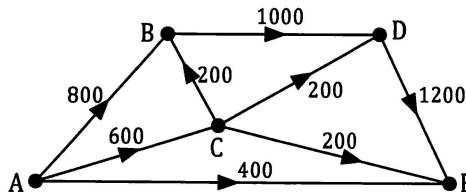
Exercise 6A. Page 119.

1. 60 units 2. 90 units 3. 80 units 4. 100 units
5. 450 units 6. 550 units 7. 500 units 8. 170 units
9. 550 units 10. 100 units
11. Max flow is 850 units.

One way it can be achieved is shown below.



12. $v = 100,$ $w = 300,$ $x = 100,$ $y = 200,$ $z = 150.$
13. 1800



14. 16 units
 Upgrade AC to 8 units: Increase the maximum flow to 19.
 Upgrading BD or FG: Will not alter the maximum flow.

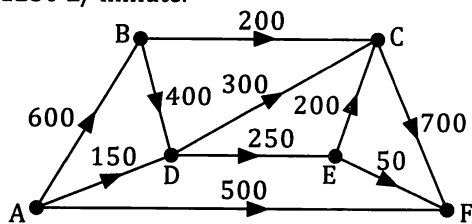
Exercise 6B. Page 122.

1. I: 150 units, II: 130 units, III: 160 units, IV: 140 units, V: 180 units.

2. 450
3. 550
4. 500
5. 170
6. 550
7. 100

Miscellaneous Exercise Six. Page 123.

1. As percentages, and to the nearest percent, the seasonal indices for the first, second, third and fourth quarters are 141%, 112%, 59% and 88% respectively.
Numbers for the first quarter tend to be 41% up on the quarterly average, for the second quarter 12% up on the quarterly average, for the third quarter 41% down on the quarterly average and for the fourth quarter 12% down on the quarterly average.
2. 241 units
3. (a) \$4026·14 needs to be repaid per month for the loan to be paid off in 25 years.
(Note: Rounding up from \$4026·135417 so final payment will be a little under \$4026·14.)
(b) A total of \$707840 (nearest \$10) interest is paid during the 25 years.
Note: $\$4026 \cdot 14 \times 12 \times 25 - \$500\,000$ gives \$707842.
Allowing for reduced 300th payment gives \$707837·27.
Different rounding regime, eg on some online amortization calculators, gives \$707837·52.
4. (a) \$12000 (b) 7·2% (c) \$18211·68
5. \$14400 in first year, \$12672 in second year, \$11151·36 in third year.
6. 1250 L/ minute.

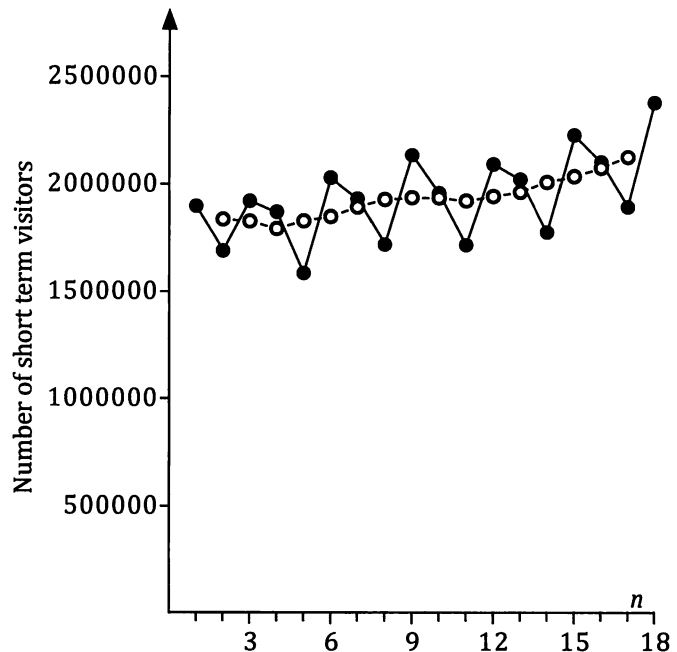


(The maximum flow is 1250 L/min but the maximum flow diagram shown here is not the only possible way of achieving it.)

7. (a) Three point moving averages are shown below. (Values are given to the nearest 100, as requested.)

n	3 pt MA (M)
1	—
2	1837500
3	1828600
4	1794000
5	1830100
6	1850500
7	1894900
8	1930000
9	1938800
10	1937900
11	1923600
12	1944700
13	1964700
14	2010700
15	2037400
16	2076800
17	2126800
18	—

(b)

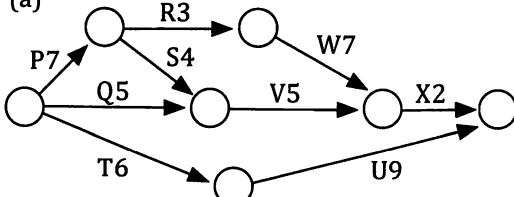
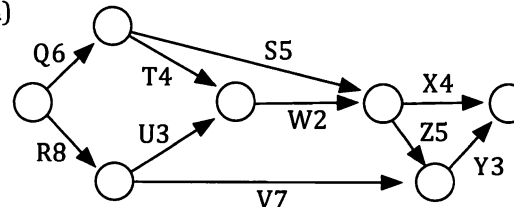
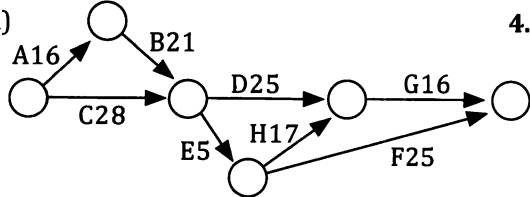
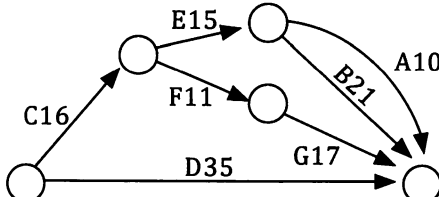
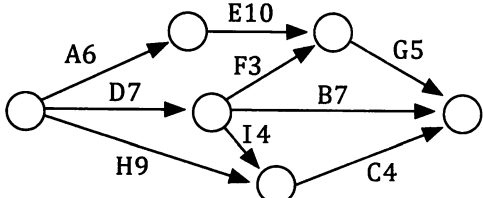
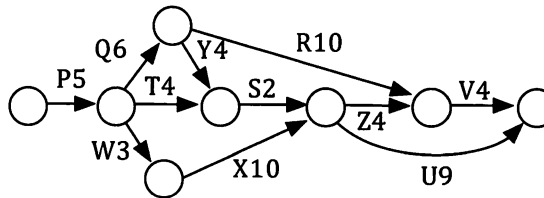
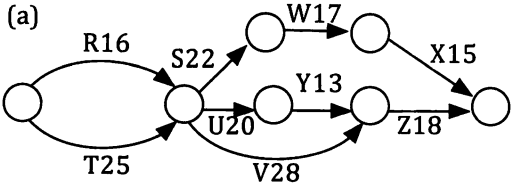
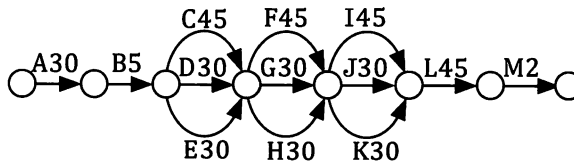
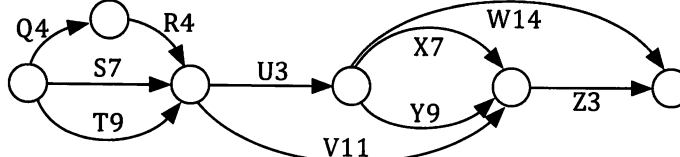
(c) $M = 18970n + 1\,753\,000$ (d) Interpretation of the number 18970 in the equation $M = 18970n + 1\,753\,000$:

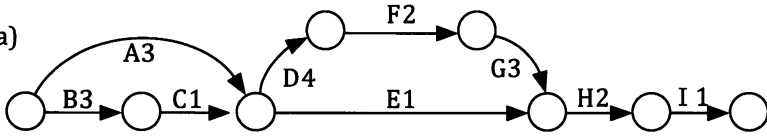
The number of short term visitors arriving in Australia from January 2008 to December 2013 was, on average, increasing at a rate of approximately 19 000 people per four months.

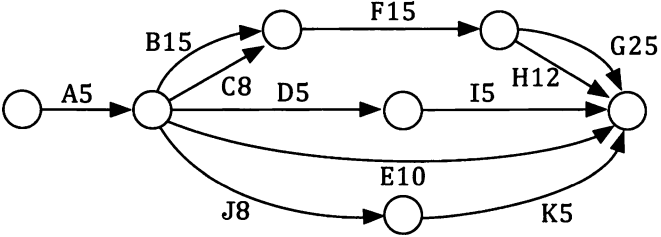
Exercise 7A. Page 128.

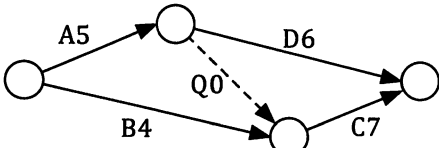
1. (a) 24 days (b) P T U W Y (c) 3
2. (a) 12 hrs (b) A E F H (c) 12 hrs (i.e. no change) (d) 1
3. (a) 17 days (b) A C H L N (c) 17 days (i.e. no change)
4. (a) 27 hours (b) P S V Z (c) No (cut by 1 hour, P W X Y becomes critical) (d) P
5. (a) 30 days (b) A C G J (c) 14 days into the project. (d) 16 days into the project.
(e) 2 days (f) 4 days (g) 3 days (h) F
6. (a) 74 hours (b) P Q R S (c) 10 hours (d) 29 hours (e) 19 hours (f) 57 hours (g) 10 hours
7. 17 minutes, P T Y and S X Z (b) 2 minutes each.
8. (a) 44 days (b) B C G J (c) 14 days (d) 2 days (e) 4 days
9. (a) 3h 10 min (b) 12.10 pm (c) 11.05 am
10. (a) 4.25 pm (b) 4.55 pm (c) 45 mins

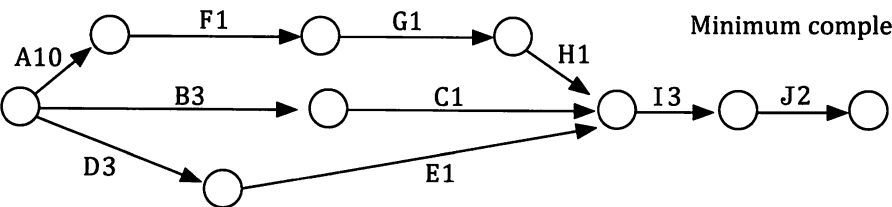
Exercise 7B. Page 135.

1. (a) 
(b) 19 weeks, P - R - W - X
(c) 4 weeks
2. (a) 
(b) 21 days, R - U - W - Z - Y
(c) 23 days
3. (a) 
(b) 78 minutes, A - B - D - G
(c) 3 minutes
4. (a) 
(b) 52 days, C - E - B
(c) 8
5. (a) 
(b) 21 hours, A - E - G
(c) D and H
6. 
27 minutes, P - W - X - U
7. (a) 
(b) 79 days, T - S - W - X
(c) 80 days
8. 
11.50 am
9. (a) 
(b) 26 hours, T - U - W
(c) (i) Two hours over the allocated three.
(ii) Three hours over the allocated three.

10. (a)  (b) 16 days
B - C - D - F - G - H - I
(c) 8 days
(d) Reduce by one day.

11.  Minimum completion time 60 minutes.
Attempt to speed up:
Service vehicles getting in position,
passengers on and off,
security check.

12.  Critical path is A - Q - C
Minimum completion time is 12 hours.

13.  Minimum completion time is 18 weeks.

Miscellaneous Exercise Seven. Page 140.

1. A seasonal index for January of 1.12 means that whatever it is that was used to determine the index, eg sales numbers, gross takings, number of people, prices etc, increases in January by 12% of the monthly average.
2. Five years after the commencement of the loan the amount owed is \$243862.70 (to the nearest cent).
3. Seasonally adjusted figures are 2208 for January and 2269 for February.
4. (a) A, C. (b) B, D. (c) A, C, D. (d) C (e) D
5. The maximum number of units that can be delivered is 750.
Increasing the capacity of AC to 250 units increases the max flow by 50 units (to 800 units).
6. Connections need to be: A to B, B to base, Base to D, D to C. \$460 000
7. The following pairs of lookout positions should have direct road connections constructed: 1 and 2, 1 and 5, 2 and 3, 3 and 4
8. The greatest number of vehicles that could arrive at Z per hour, having come from A, is 2700.

Exercise 8A. Page 147.

1. Firm A to do load 2.
Firm B to do load 1.
(Total cost \$240.)
2. Firm A to do load 1.
Firm B to do load 3.
Firm C to do load 2.
(Total cost \$340.)
3. Firm A to do load 2.
Firm B to do load 4.
Firm C to do load 3.
Firm D to do load 1.
(Total cost \$3100.)

4. Operator A on machine 1.
Operator B on machine 2.
(Total number of components 180.)
5. Operator A on machine 2.
Operator B on machine 1.
Operator C on machine 3.
(Total number of components 610.)
6. Operator A on machine 2.
Operator B on machine 1.
Operator C on machine 4.
Operator D on machine 3.
(Total number of components 375.)

Exercise 8B. Page 154.

1. Company A to task 2.
Company B to task 1. Total cost 177 units.
2. Company A to task 2.
Company B to task 1.
Company C to task 3. Total cost 300 units.
3. Company A to task 3.
Company B to task 1.
Company C to task 4.
Company D to task 2. Total cost 43 units.
4. Company A to task 4.
Company B to task 5.
Company C to task 2.
Company D to task 6.
Company E to task 1.
Company F to task 3. Total cost 135 units.
5. Company A to task 2. Benefit 120 units.
Company B to task 1. Benefit 240 units.
6. Company A to task 1. Benefit 29 units.
Company B to task 3. Benefit 28 units.
Company C to task 2. Benefit 33 units.
Or
Company A to task 3. Benefit 31 units.
Company B to task 1. Benefit 26 units.
Company C to task 2. Benefit 33 units
7. Company A to task 2. Benefit 240 units.
Company B to task 3. Benefit 140 units.
Company C to task 4. Benefit 610 units.
Company 4 to task 1. Benefit 160 units.
8. Company A to task 2. Benefit 40 units.
Company B to task 1. Benefit 51 units.
Company C to task 3. Benefit 42 units.
9. Company A to task 3. Benefit 33 units.
Company B to task 4. Benefit 43 units.
Company C to task 1. Benefit 35 units.
Company D to task 2. Benefit 45 units.

10. (b) Steps of algorithm not shown here. Optimal solution is as follows:
 Adelaide rep oversees Sydney office. (Return airfare \$380.)
 Brisbane rep oversees Darwin office. (Return airfare \$570.)
 Hobart rep oversees Melbourne office. (Return airfare \$290.)
 Perth rep oversees Broome office. (Return airfare \$720.)
 (Total of return airfares is then \$1960.)
11. (a) Quick Courier Co. to job 1. (\$290)
 Speedy Courier Co. to job 3. (\$260)
 Deliverit Couriers to job 2. (\$80)
 Competitive Couriers to job 4. (\$140)
 Total cost of this minimum total cost solution is \$770.
 Or
 Quick Courier Co. to job 4. (\$150)
 Speedy Courier Co. to job 3. (\$260)
 Deliverit Couriers to job 2. (\$80)
 Competitive Couriers to job 1. (\$280)
 Total cost of this minimum total cost solution is \$770.
- (b) If all four companies add \$50 for job 4 the optimum allocation would not change, i.e. the two alternatives would still be as above (but the total cost would increase to \$820).
- (c) If all four companies multiplied their prices for job 2 by 4 the optimum situation would change.
 Quick Courier Co. would do job 1.
 Speedy Courier Co. would still do job 3.
 Deliverit Couriers would do job 4.
 Competitive Couriers would do 2.
 (Total cost of this minimum total cost solution now \$970.)
- (d) Competitive couriers. (Total \$730.)
12. Jack to area 1. Jim to area 4. Judy to area 3.
 Nicci to area 5. Ti to area 2. Total sales 50.
13. Ashok to location 3. Daniel to location 1. Kirsten to location 2. Shani to location 4.
14. Two possibilities:
 A → 1 (40 km) B → 5 (30 km) C → 2 (30 km) D → 3 (40 km) E → 4 (40 km).
 Or
 A → 3 (30 km) B → 5 (30 km) C → 2 (30 km) D → 1 (50 km) E → 4 (40 km).
 (Each possibility gives a total distance travelled of 180 km.)
15. Factory A to manufacture the Charger.
 Factory B to manufacture the Exeon.
 Factory C to manufacture the Bijou.
 Factory D to manufacture the Devine.
 Factory E to manufacture the Arcain.

Exercise 8C. Page 162.

1. Assign Alex to task 3 (15 minutes)
 Assign Ben to task 4 (17 minutes)
 Assign Theo to task 1 (21 minutes)
 Working simultaneously all three will have finished after 21 minutes.
 Yes, if all working together they could finish three tasks in 20 minutes if Alex assigned to task 4 (20 minutes), Ben to task 1 (18 minutes) and Theo to task 3 (18 minutes). Working simultaneously the three tasks would be completed after 20 minutes.

2. Assign the digger from the Horton storage facility to Abbysix. (25 km)
Assign the digger from the Peel storage facility to Crampon. (18 km)
Assign the digger from Wimsley storage facility to Denton. (31 km)
Total distance 74 km.
3. Assign taxi A to Tenielle, who will have to wait 8 minutes.
Assign taxi C to Zarani, who will have to wait 6 minutes.
Assign taxi D to Bill who will have to wait 8 minutes.
4. Assign person B to job 2. (\$220)
Assign person C to job 1. (\$120)
Assign person D to job 3. (\$160)
Assign person E to job 4. (\$150)
Total profit is \$650.

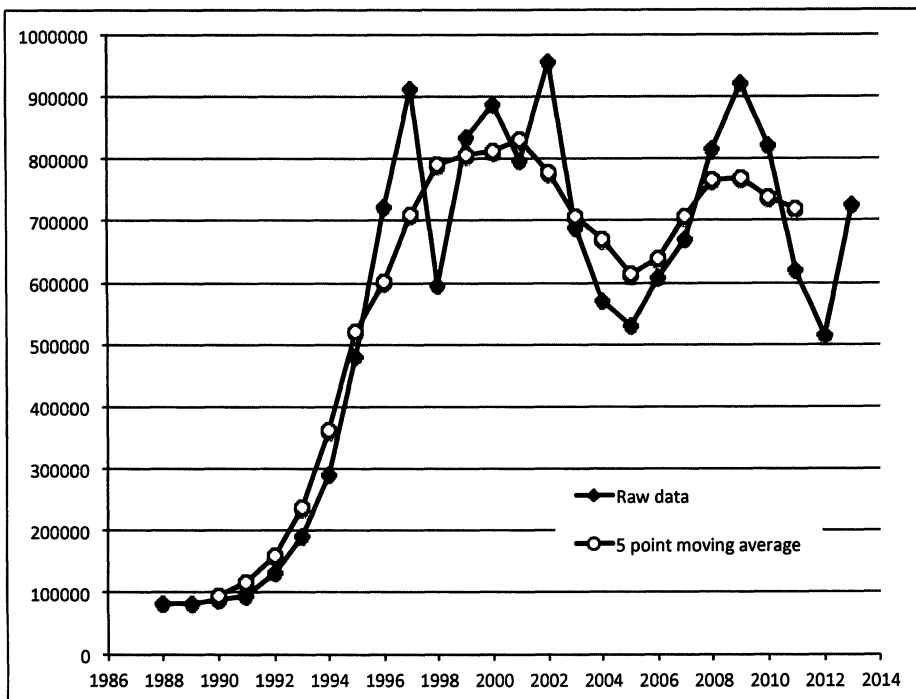
Challenge. Page 164.

Optimum solution is	Worker A to task 9.	15 minutes.	
	Worker B to task 6.	15 minutes.	
	Worker C to task 4.	21 minutes.	
	Worker D to task 7.	29 minutes.	
	Worker E to task 3.	9 minutes.	
	Worker F to task 10	15 minutes	
	Worker G to task 1.	17 minutes.	
	Worker H to task 2.	20 minutes.	
	Worker I to task 5.	14 minutes.	
	Worker J to task 8.	36 minutes.	Total 191 minutes.

Miscellaneous Exercise Eight. Page 165.

- | 1. | Compounding frequency. | Nominal annual interest rate (%). | Effective annual interest rate (%). |
|----|------------------------|-----------------------------------|-------------------------------------|
| | Annual | 10% | 10% |
| | Six monthly | 10% | 10.25% |
| | Quarterly | 10% | 10.38% (2dp) |
| | Monthly | 10% | 10.47% (2dp) |
| | Weekly | 10% | 10.51% (2dp) |
| | Daily | 10% | 10.52% (2dp) |
2. (a) \$553 382 (b) \$553 382
 3. Correct answer is \$662 642, to the nearest dollar.
 4. Immediately after the fifteenth repayment has been made the amount owed is \$4908.28 (to the nearest cent).
 5. 27 hours, B-E-J-M
 6. (a) The greatest number of messages that can go from A to G in an hour is 850.
(b) The greatest number of messages that can go from A to D in an hour is 650.
 7. The maximum flow from A to X is 1200 units.
The extra capacity along EX increases the maximum flow from A to X by 150 units.
 8. Fran should swim backstroke (54.2 seconds).
Marta should swim breaststroke (48.2 seconds).
Julie should swim butterfly (43.6 seconds).
Micah should swim freestyle (42.4 seconds).
Total time 188.4 seconds.

9.

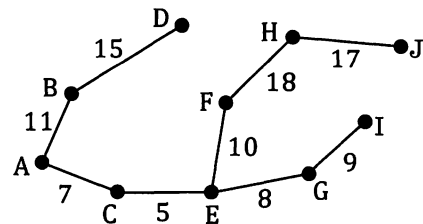


Compare your comments describing any trends in the numbers of live cattle exported from Australia per year over the period of time from 1988 to 2013 with those of others in your class.

10. Vehicle A to location 3 7 minutes.
 Vehicle B to location 2 9 minutes.
 Vehicle C to location 1 6 minutes.
 Vehicle D to location 5 9 minutes.
 Vehicle E to location 4 10 minutes.

11. (a) 6850
 (b) 1st Quarter 4932
 2nd Quarter 6028
 3rd Quarter 7261
 4th Quarter 9179

12. The total length of the minimum spanning tree for the given network is 100 units.
 The minimum span is shown on the right.



13. (a) 25 days, A - C - F - G - J
 (b) 3 days
14. (Answers may vary slightly dependent on level of rounding used when calculating.)

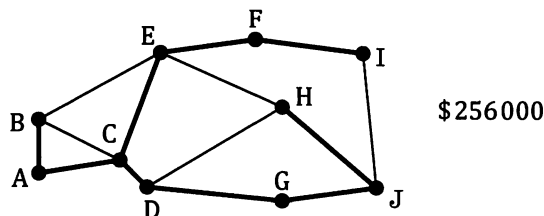
- (a) 1st two months index 0.3579 2nd two months index 0.5675
 3rd two months index 1.6293 4th two months index 1.8760
 5th two months index 1.1878 6th two months index 0.3815

(b)	1st two months	2nd two months	3rd two months	4th two months	5th two months	6th two months
Year 1	3460	3190	3120	3370	3180	2760
Year 2	3860	3870	3940	3810	4100	4280
Year 3	4200	4560	4580	4380	4310	4690

- (c) Viewing the graph indicates linear modelling is suitable. (It could be argued that the sixth deseasonalised value is an outlier and hence exclude it from the calculation of the regression line but answers given here are based on the inclusion of all points.)
- (d) $D = 95.52n + 2960$
- (e) 1st two months of year 4, $n = 19$, estimated $N = 1710$ (nearest ten).
4th two months of year 4, $n = 22$, estimated $N = 9500$ (nearest ten).
- (f) Compare your possible events with those of others in your class.

15. (a) 21 weeks, A-D-H-I-J-L
(b) 11 weeks
(c) 19 weeks

16.



Direct connection from H to I is worth doing if it can be done for less than \$38 000 as it could then replace EC in the minimum span.

17. Either assign Machine A to product 3 (3 300 units)
Machine B to product 4 (4 100 units)
Machine C to product 1 (4 000 units).
OR assign Machine A to product 4 (3 300 units)
Machine B to product 3 (4 100 units)
Machine C to product 1 (4 000 units).

(Either way gives the total number of units produced as 11400.)

18. (a) 1155 m (b) Goods In - Dock 1, Customs - Goods Out, Customs - Waiting,
Customs - Dock 3, Dock 1 - Dock 2, Dock 2 - Customs.
19. Allocate DG Transport to Job 4. (\$16 500.)
Allocate Haulage and Co. to Job 2. (\$12 800.)
Allocate JA Movers to Job 1. (\$18 000.)
Allocate Move it and Shift it to Job 3. (\$25 000.)
Total cost \$72 300.